

# Customer Churn Prediction

*A machine-learning system that flags at-risk customers before they leave*

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## THE BUSINESS PROBLEM

# Losing a customer costs far more than keeping one

Churn is one of the most expensive problems in a subscription business. Every customer who leaves erases the cost it took to acquire them — and retention teams can only act if they know who is at risk before it happens.

### The question this project answers:

*“Given what we already know about a customer, can we predict — before they leave — whether they're at risk of churning?”*



1,000

customer records analyzed



88.3%

observed churn rate in the sample



4

core signals used to score every customer

## THE DATA

# What I built the model on



**1,000**

### CUSTOMER RECORDS

Combining demographics, account details, and billing history



**10**

### DATA ATTRIBUTES

Age, gender, tenure, charges, contract type, support & more



**297**

### VALUES FILLED IN

Missing InternetService entries handled before modeling



**0**

### DUPLICATE RECORDS

Full 1,000-row dataset verified clean and ready to use

*Feature set used by the model: Age · Gender · Tenure · Monthly Charges*

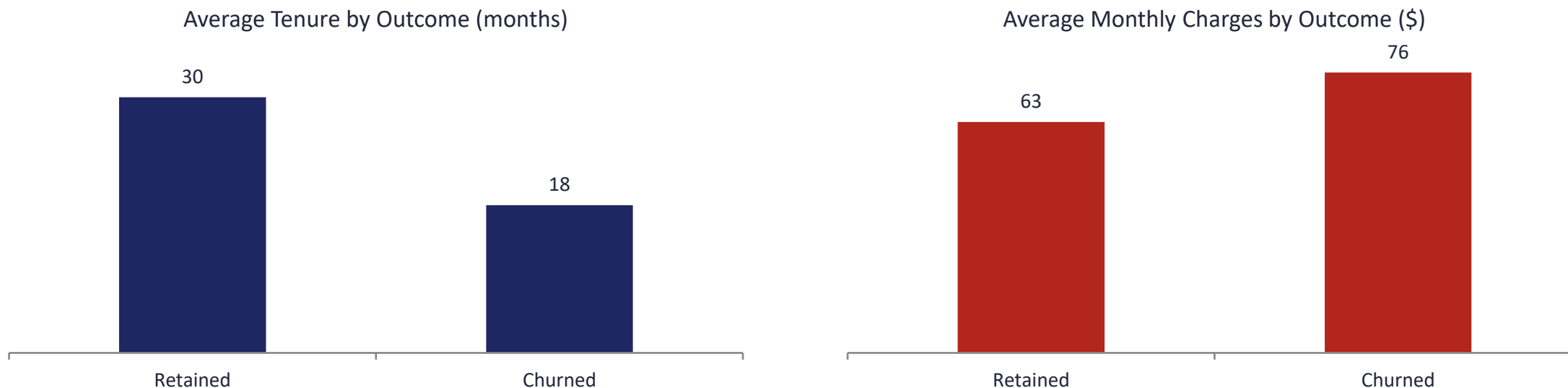
## CLASS BALANCE — CHURN LABEL

**88.3% Churned (883)**

**11.7% Retained (117)**

## KEY INSIGHT

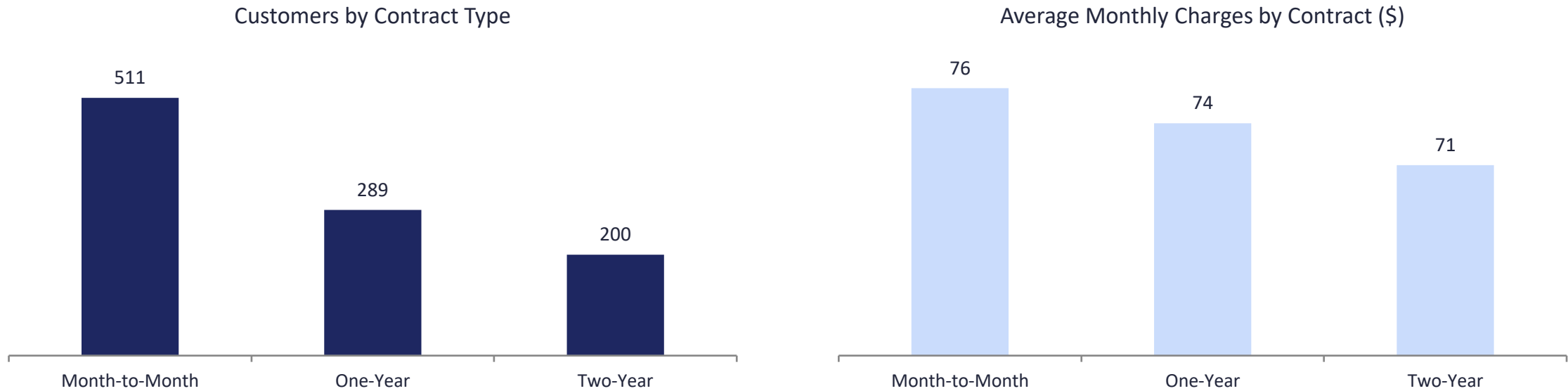
# Tenure and price are the clearest churn signals



- Customers who churn have an average tenure of just 17.5 months — almost half that of customers who stay (30.3 months). Newer customers are the highest-risk group.
- Churned customers pay about \$13/month more on average (\$75.96 vs. \$62.55) — higher bills are linked to a higher chance of leaving.
- Age shows almost no difference between groups (44.8 vs. 43.5) — it is not a meaningful churn driver on its own.

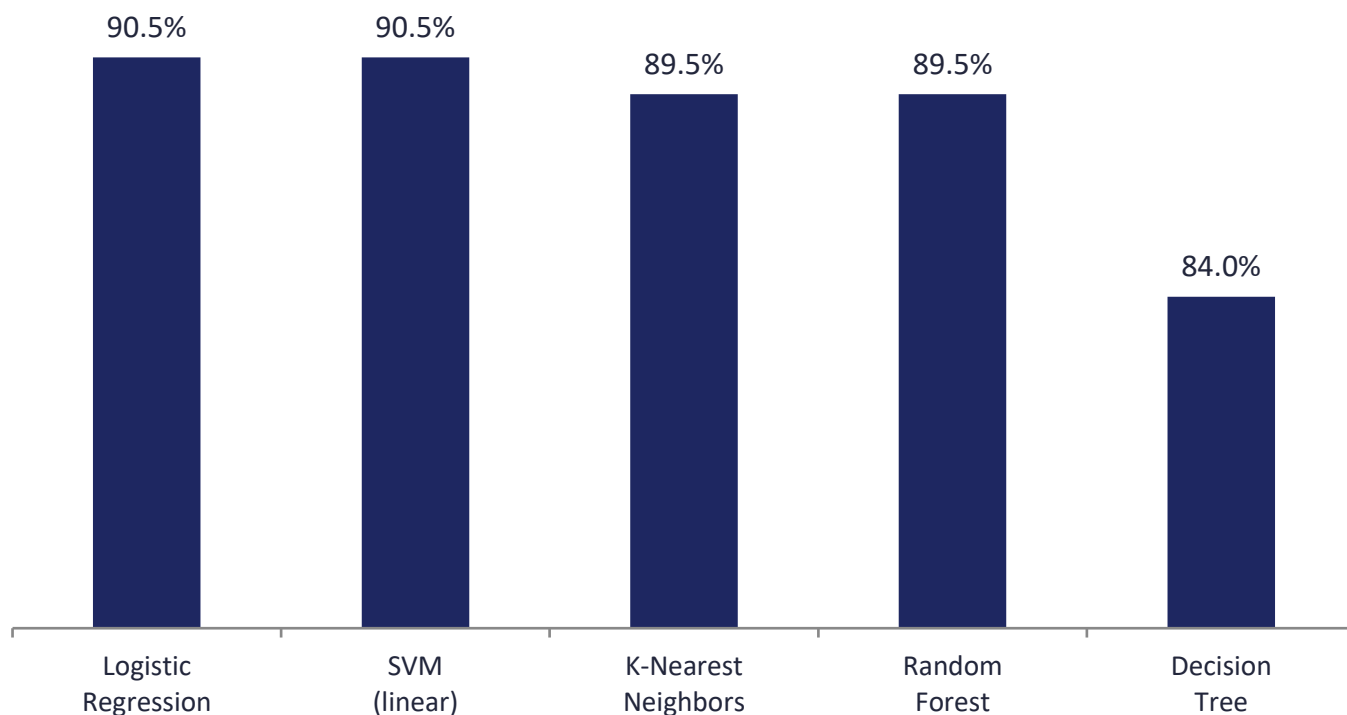
KEY INSIGHT

# Longer contracts trend toward lower bills



Over half of customers (511 of 1,000) are on flexible Month-to-Month plans — the same segment most exposed to churn. Average charges step down slightly with longer commitments, suggesting a loyalty-discount effect worth reinforcing through retention offers on the Month-to-Month base.

# Five models tested — two tied for the top spot



## HOW EACH MODEL WAS TUNED

**Logistic Regression**

*Baseline*

**SVM (linear)**

*Grid search — selected model*

**K-Nearest Neighbors**

*Grid search, k=5*

**Random Forest**

*Grid search, 128 trees*

**Decision Tree**

*Grid search, tuned depth*

# Why I chose a linear Support Vector Machine



## Support Vector Machine

*linear kernel ·  $C = 0.01$*

# 90.5%

test-set accuracy — tied for the best result across all five models tested



### Simple, stable boundary

Strong regularization favors a wide margin that generalizes well on a modest, 1,000-record dataset.



### Easy to explain

A linear decision boundary means churn risk moves smoothly and predictably with each input — no hidden complexity.



### No accuracy trade-off

It matched every other model on test accuracy, so we kept the simpler, more transparent option.



**Inputs: Age · Gender · Tenure · Monthly Charges → scaled with the matching scaler → Prediction: Churn or Not Churn**

**Thank you**